

# SelectionBias: An R Package for Bounding Selection Bias in Causal Estimands

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**Abstract** Selection bias may arise when there are dropouts or missing data in the analysis, or when subjects are included or excluded in the analysis based upon some selection criteria for the study population. Selection bias can jeopardize the validity of the study and a sensitivity analysis for assessing the effect of the selection is desired. Recently, there has been a surge of results for selection bias in causal inference, with several suggestions for sensitivity analyses. One method is to construct bounds for the bias. Here, we present the R package SelectionBias that can be used to calculate previously proposed bounds for selection bias for the causal risk ratio and causal risk difference for both the total and the selected populations. The first bound, derived by Smith and VanderWeele (SV), is based on values of sensitivity parameters that describe parts of the joint distribution of the outcome, treatment, selection indicator and unobserved variables. The second bound is an improved sharp bound that uses the same sensitivity parameters as the SV bound. The third bound is based solely on the observed data, and is therefore referred to as the assumption-free (AF) bound. The fourth and fifth bounds, the generalized assumption-free (GAF) and counterfactual assumption-free (CAF) bounds, utilize both the data and sensitivity parameters. The R package is illustrated with a simulated dataset that emulates a study where the effect of the zika virus on microcephaly in Brazil is investigated. Lastly, its performance and features are compared to the already existing R package EValue, highlighting situations where the two packages provide distinct advantages over each other.

## 1 Introduction

Selecting a study population from a larger source population is a common procedure, for example in an observational study with data from a population register. Based on the research question, the study population is commonly constructed using inclusion/exclusion criteria. The construction of the study population from these selection criteria might introduce bias when estimating the causal effect of the treatment on the outcome of interest (Hernán et al., 2004). This systematic error is commonly referred to as selection bias and can occur both when interest lies in the total population and in the selected subpopulation. Selection bias can also arise if, for example, there are dropouts or other missing values for some individuals in the study (Lu et al., 2022). In an empirical study, it is often of interest to assess the magnitude of potential biases using a sensitivity analysis. One type of sensitivity analysis is calculating bounds for the bias, see e.g. Huang and Lee (2015), Greenland (2003), Flanders and Ye (2019), Smith and VanderWeele (2019), Zetterstrom and Waernbaum (2022), Zetterstrom (2024), Zetterstrom et al. (2025), and Duarte et al. (2024) for selection bias.

This paper presents the R package SelectionBias which can be used to calculate different bounds for selection bias. The ability to calculate different bounds is attractive as these bounds are in many cases based on assumptions that may or may not be fulfilled. One of the bounds included in SelectionBias is the bound presented in Smith and VanderWeele (2019), hereafter referred to as the SV bound. This bound is already included in the R package EValue and can also be calculated using an online calculator (Smith and VanderWeele, 2019; Smith et al., 2021). For these software tools, the user inputs assumed values of sensitivity parameters, which are summary measures of a joint distribution. Thus, these sensitivity parameters can be difficult to specify, especially in the presence of many selection variables. In SelectionBias, we also allow the user to input the model, which might be easier to specify than the sensitivity parameters. SelectionBias also allows the user to assess whether the SV bound is sharp. The sharpness of a bound is a desirable property, and recently proposed sharp bounds based on the same sensitivity parameters as the SV bound (Zetterstrom et al., 2025) are also included in SelectionBias. These bounds are referred to as the sharp bounds. SelectionBias also includes functions for calculating the AF, GAF, and CAF bounds (Zetterstrom and Waernbaum, 2022; Zetterstrom, 2024). The sharp, AF, GAF, and CAF bounds cannot be calculated using other packages available online. Notably, all the bounds in the SelectionBias package are calculated within strata of the pre-treatment covariates. The contents of the SelectionBias package is:

- `zika_learner`: a simulated dataset inspired by the study in de Araújo et al. (2018) and the zika example in Smith and VanderWeele (2019). The dataset includes seven variables: zika virus, microcephaly, the selection variables birth and public hospital, the selection indicator variable and the two “unmeasured” variables living area and socioeconomic status.
- `sensitivityparametersM()`: a function that calculates the sensitivity parameters for the SV, sharp, and GAF bounds for a model following the generalized M-structure (Figure 1) defined

by the user.

- `SVbound()`: a function that calculates the SV lower and upper bound for the risk ratio or risk difference in either the total or subpopulation for sensitivity parameters given by the user, or calculated from `sensitivityparametersM()`.
- `sharpbound()`: a function that calculates the sharp lower and upper bound for the risk ratio or risk difference in either the total or subpopulation for sensitivity parameters given by the user, or calculated from `sensitivityparametersM()`.
- `AFbound()`: a function that calculates the AF lower and upper bounds for the risk ratio or risk difference in either the total or subpopulation. The input can either be a dataset or probabilities calculated from the data.
- `GAFbound()`: a function that calculates the GAF lower and upper bounds for the risk ratio or risk difference in either the total or subpopulation for a dataset or probabilities from the data, together with sensitivity parameters either given by the user, or calculated from `sensitivityparametersM()`.
- `CAFbound()`: a function that calculates the CAF lower and upper bounds for the risk ratio or risk difference in either the total or subpopulation for a dataset or probabilities from the data, together with sensitivity parameters given by the user.
- `checksharpSVbound()`: a function that evaluates if the SV bound is sharp.

The paper is outlined as follows. In Section 2, we present the causal framework, estimands and corresponding selection biases together with an introduction of the bounds. In Section 3, the simulated example dataset `zika_learner` is described. In Section 4, the R package `SelectionBias` is demonstrated using the `zika_learner` data. The performance and features of `SelectionBias` are compared to other R packages for sensitivity analyses, especially `EValue`, in Section 5. Finally, the paper is summarized in Section 6.

## 2 Causal framework and selection bias

In this section, we provide a theoretical background for the bounds calculated in the R package `SelectionBias`. First, we introduce the notation and causal framework, and second, we describe the SV, sharp, AF, GAF, and CAF bounds.

We assume  $K$  binary selection variables,  $S_1, \dots, S_k, \dots, S_K$ , where  $S_k$  indicates if the subject passes the  $k$ th selection criterion, from which we define a selection indicator variable  $I_S$  such that

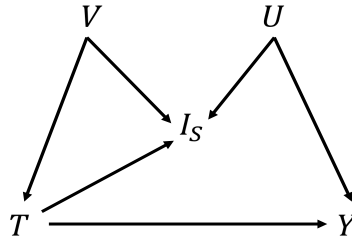
$$I_S = \begin{cases} 1 & \text{if } \prod_{k=1}^K S_k = 1 \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

For a subject to be included in the study, the corresponding  $I_S$  must be equal to 1. We consider an i.i.d. sample of size  $i = 1, \dots, n$  units from the subpopulation such that  $I_S = 1$ , but henceforth suppress the index  $i$  representing units in the sample. We also assume a binary treatment,  $T = 1$  if the unit is treated and  $T = 0$  if the unit is not treated, and two corresponding binary potential outcomes,  $Y(1)$  and  $Y(0)$  as well as an observed outcome such that  $Y = TY(1) + (1 - T)Y(0)$ , i.e. consistency is assumed (Rubin, 1974). Additionally, we assume a vector of observed pre-treatment covariates, denoted by  $X$ , such that conditional exchangeability holds in the total population:  $Y(t) \perp\!\!\!\perp T \mid X$ ,  $t = 0, 1$ . However, similar to the previous literature, we suppress  $X$  throughout to make the notation simpler and assume that all calculations are performed within strata of the pre-treatment covariates. Lastly, we define a vector of unobserved pre-treatment covariates,  $U$ , which is part of the assumptions in the sensitivity analysis with the SV, sharp, and GAF bounds. It is illustrated in the generalized M-structure (Figure 1), where  $U$  is a predictor of the outcome. In contrast, the unobserved covariates  $U$  are not needed in the AF and CAF bounds, which impose fewer assumptions since dependencies with  $U$  are not needed.

The causal estimands of interest are the causal risk ratio,  $\beta_R = P(Y(1) = 1)/P(Y(0) = 1)$ , and causal risk difference,  $\beta_D = P(Y(1) = 1) - P(Y(0) = 1)$ , in the total population, and the causal risk ratio,  $\beta_{R_S} = P(Y(1) = 1|I_S = 1)/P(Y(0) = 1|I_S = 1)$ , and the causal risk difference,  $\beta_{D_S} = P(Y(1) = 1|I_S = 1) - P(Y(0) = 1|I_S = 1)$ , in the subpopulation. We also define the observed risk ratio and risk difference under selection,  $I_S = 1$ , as  $\beta_R^{obs} = P(Y = 1|T = 1, I_S = 1)/P(Y = 1|T = 0, I_S = 1)$  and  $\beta_D^{obs} = P(Y = 1|T = 1, I_S = 1) - P(Y = 1|T = 0, I_S = 1)$ . These are estimated from data, but we assume that there is no sampling variability and that  $\beta_R^{obs}$  and  $\beta_D^{obs}$  are population quantities. The selection bias for the risk ratio estimands,  $\beta_R$  and  $\beta_{R_S}$ , are defined as the ratio between the risk ratio and the causal estimand,

$$Bias(\beta_R) = \beta_R^{obs} / \beta_R, \quad (2)$$

$$Bias(\beta_{R_S}) = \beta_R^{obs} / \beta_{R_S}, \quad (3)$$



**Figure 1:** The generalized M-structure relating treatment, outcome, and selection to two unmeasured variables. Nodes represent the treatment ( $T$ ), outcome ( $Y$ ), selection indicator ( $I_S$ ), and two unmeasured pre-treatment variables ( $V$  and  $U$ ); arrows represent possible direct effects between the connected variables, with  $V$  pointing to both  $T$  and  $I_S$ , and  $U$  pointing to both  $I_S$  and  $Y$ . This structure satisfies the conditional independence assumptions required for the SV, sharp, and GAF bounds, since  $Y$  does not have a direct path to  $I_S$  that bypasses the other variables.

and the selection bias for the risk difference estimands,  $\beta_D$  and  $\beta_{D_S}$ , are defined as the difference between the observed risk difference and the causal estimand,

$$\text{Bias}(\beta_D) = \beta_D^{obs} - \beta_D, \quad (4)$$

$$\text{Bias}(\beta_{D_S}) = \beta_{D_S}^{obs} - \beta_{D_S}. \quad (5)$$

When the observed risk ratio or risk difference do not coincide with the causal estimands, i.e. when selection bias is present, bounds for the causal estimands can be calculated and compared to the observed quantity in order to assess the magnitude of the selection bias. We present a package that can be used to perform sensitivity analysis for the selection bias, under various assumptions. Bounds from the following works are considered; [Smith and VanderWeele \(2019\)](#), [Zetterstrom and Waernbaum \(2022\)](#), [Zetterstrom \(2024\)](#), and [Zetterstrom et al. \(2025\)](#). The ideas behind the bounds are shortly described here with a summary of their properties in Table 1. For completeness, all bounds are presented in the Appendix.

The SV bounds rely on two versions of conditional independence assumptions involving the outcome, selection indicator, treatment, and a vector of unmeasured variables (see Appendix). The assumptions differ depending on the estimand of interest. If total population estimands  $\beta_R$  and  $\beta_D$  are of interest, conditioning on  $U$  and  $T$  must be sufficient for conditional independence of the observed outcome  $Y$  and the selection indicator,  $I_S$ . For the case when subpopulation estimands  $\beta_{R_S}$  and  $\beta_{D_S}$  are considered, an assumption requiring exchangeability conditional on  $U$  and  $I_S$  is instead made. An example where both assumptions are fulfilled is the M-structure, Figure 1, although other structures are possible. The purpose of the different assumptions is to describe a setting where unbiased estimates could be obtained if the unmeasured variables were observed. The SV bounds are based on sensitivity parameters that are constructed as risk ratios formed by the joint distribution of the outcome, treatment, selection indicator variable and unmeasured variables,  $(Y, T, I_S, U)$ . The sensitivity parameters are

$$\text{RR}_{SU|ts} = \max_u \frac{p(U = u \mid I_S = s, T = t)}{p(U = u \mid I_S = 1 - s, T = t)},$$

$$\text{RR}_{UY|t} = \frac{\max_u p(Y = 1 \mid U = u, T = t)}{\min_u p(Y = 1 \mid U = u, T = t)},$$

and

$$\text{BF}_{ts} = \frac{\text{RR}_{SU|ts} \times \text{RR}_{UY|t}}{\text{RR}_{SU|ts} + \text{RR}_{UY|t} - 1},$$

from which bounds for the causal risk ratio and causal risk difference can be calculated. For the subpopulation, SV defined the sensitivity parameters

$$\text{RR}_{TU|t} = \max_u \frac{p(U = u \mid I_S = 1, T = t)}{p(U = u \mid I_S = 1, T = 1 - t)},$$

$$\text{RR}_{UY|S=1} = \max_t \frac{\max_u p(Y = 1 \mid U = u, T = t, I_S = 1)}{\min_u p(Y = 1 \mid U = u, T = t, I_S = 1)},$$

and

$$BF_t = \frac{RR_{TU|t} \times RR_{UY|S=1}}{RR_{TU|t} + RR_{UY|S=1} - 1}.$$

More details of the sensitivity parameters and bounds can be found in the original work. It may be difficult to interpret and specify the sensitivity parameters, especially in the presence of multiple selection variables and multiple unmeasured variables. Instead, it may be easier to break down the bound into smaller pieces, and calculate the sensitivity parameters for a specified data generating process (DGP). We facilitate such calculations in the R package [SelectionBias](#).

A bound is sharp if the bias can equal the value of the bound, given the necessary assumptions, an observed distribution, and correctly specified sensitivity parameters. Since different bounds require different assumptions and information, two bounds for the same causal estimand can be simultaneously sharp if each is valid under its own requirements. The SV bounds are only sharp under certain conditions, i.e. sometimes the SV bounds are too conservative and are larger than the bias can actually be. As an alternative, the sensitivity parameters have been used to construct improved bounds that are generally sharp ([Zetterstrom et al., 2025](#)), referred to as the sharp bounds. However, it should be noted that they do in some cases require some additional knowledge of the selection probability.

The AF bounds are constructed solely from the data and have the advantage that no additional assumptions regarding the causal model are needed, only that  $P(I_S = 1)$  is known. The bounds are constructed as the minimum and maximum possible values of the causal estimand,  $\beta^{min}$  and  $\beta^{max}$ . These values are then used as lower and upper bounds for the causal estimand. As the bounds also demonstrate the maximum selection bias, they can be used to assess whether other bounds are informative: if another bound is greater than the AF bound, it is not informative. However, if the treatment or outcome is rare, the AF bounds for the risk ratios can be very wide, and practically non-informative. When this is the case, and additional knowledge is available, it is instead advisable to use other bounds that take that knowledge into account.

The GAF bounds are similar to the AF bounds in so far as they also utilize the observed data, but they also rely on the same conditional independence assumptions as the SV bound (see Appendix), which results in tighter bounds compared to the AF bounds. The sensitivity parameters are probabilities instead of risk ratios, which can in some instances be easier to specify. The sensitivity parameters are defined as

$$m_T = \min_{t,u} P(Y = 1|T = t, U = u)$$

and

$$M_T = \max_{t,u} P(Y = 1|T = t, U = u)$$

for the total population, and

$$m_S = \min_{t,u} P(Y = 1|T = t, U = u, S = 1)$$

and

$$M_S = \max_{t,u} P(Y = 1|T = t, U = u, S = 1)$$

for the subpopulation.

The CAF bounds are an alternative when the conditional independence assumptions are not fulfilled, but they require the user to specify counterfactual probabilities,  $P(Y(t) = 1|T = 1 - t, I_S = 1)$ . The sensitivity parameters are defined as

$$m'_T = \min_t P(Y = 1|T = t, I_S = 0)$$

and

$$M'_T = \max_t P(Y = 1|T = t, I_S = 0),$$

for the total population, and

$$m'_S = \min_t P(Y(t) = 1|T = 1 - t, I_S = 1)$$

and

$$M'_S = \max_t P(Y(t) = 1|T = 1 - t, I_S = 1),$$

for the subpopulation. The AF bounds can of course be used if the conditional independence assumption is not fulfilled, but the CAF bounds are less conservative. The GAF and CAF bounds are equal to the AF bound when the most extreme values on sensitivity parameters are used.

**Table 1:** Summary of the different bounds.

| Bound                                | Unknown sensitivity parameters | Uses data | Additional assumptions | Specify counterfactual probabilities |
|--------------------------------------|--------------------------------|-----------|------------------------|--------------------------------------|
| Smith and VanderWeele (SV)           | ✓                              |           | ✓                      |                                      |
| Sharp                                | ✓                              | ✓         | ✓                      |                                      |
| Assumption-free (AF)                 |                                | ✓         |                        |                                      |
| Generalized                          | ✓                              | ✓         | ✓                      |                                      |
| assumption-free (GAF)                |                                |           |                        |                                      |
| Counterfactual assumption-free (CAF) | ✓                              | ✓         |                        | ✓                                    |

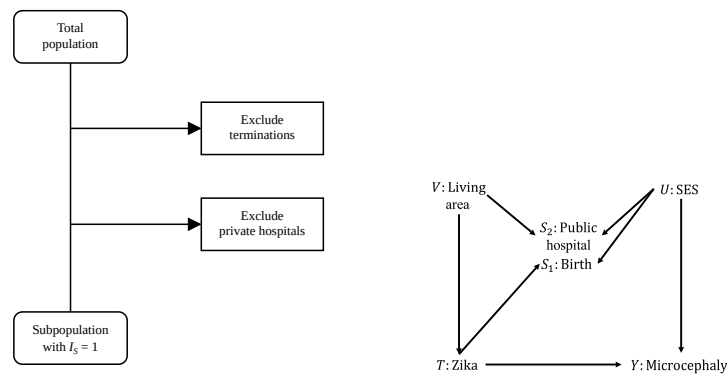
### 3 The simulated data set

For the purpose of illustration of the bounds, we construct a simulated dataset, `zika_learner`. This is inspired by a case-control study that investigates the effect of zika virus on microcephaly (de Araújo et al., 2018) and the numerical zika example introduced in Smith and VanderWeele (2019). The data can be loaded by

```
data("zika_learner", package = "SelectionBias")
```

The zika example in Smith and VanderWeele (2019) covers the case of a single selection, and we expand the example with a second selection variable to demonstrate multiple inclusion criteria which are common in empirical applications. The two selections are the binary variables `birth` and `public_hospital`, and the selection process is described in Figure 2 (left). In the original study, pre-treatment covariates for both the mother and the infant were included in order to control for confounding (de Araújo et al., 2018), however, we only consider one stratum of the covariates and thus exclude all observed pre-treatment covariates in the simulated dataset. The causal model of the dataset is given in Figure 2 (right). All variables are binary and generated from the binomial distribution such that the prevalences of the variables, and strengths of dependencies between them, mimic the real empirical data and the assumed values for the sensitivity parameters in Smith and VanderWeele (2019). The causal dependencies are generated by the logistic models described in Table 2. Even though the dataset is inspired by a case-control study, the simulated dataset emulates a register with 5000 observations, similar to an observational study of zika virus (Lebov et al., 2019). The variables included are:

- `urban` ( $V$ ). A binary, unobserved variable indicating whether the subject lives in an urban area ( $V = 1$ ) or not ( $V = 0$ ). The probability of living in an urban area is set to 0.85 following numbers from the World Bank (World Bank Group).
- `ses` ( $U$ ). A binary variable indicating socioeconomic status (high vs low), with probability set to 0.5.
- `zika` ( $T$ ). In 2016 there were approximately 34000 possible cases of zika virus during pregnancy (de Oliveira et al., 2017), and there are approximately 2.9 million births per year in Brazil (Diniz et al., 2019). The risk of getting zika was higher in urban areas (Ali et al., 2017). Thus, the prevalence of zika in the simulated dataset is around 1% with a positive impact of  $V$ , see Table 2 for details.
- `mic_ceph` ( $Y$ ). The estimated prevalence of microcephaly was 74 cases per 10000 births and the controlled odds ratio of zika virus on microcephaly was estimated at 73.1 (de Araújo et al., 2018). We mimic these values with an overall prevalence of 0.8% and a risk ratio of 74.5 among the subpopulation after two selections, see Table 2 for details.
- `birth` ( $S_1$ ). Pregnancies that ended in a live or still birth were included in the study, and pregnancies that ended in a miscarriage or an abortion were excluded. The probability of a pregnancy ending in birth is assumed to be affected both by zika virus infection and socioeconomic status. The number of births per year in Brazil is approximately 2.9 million and the number of unregulated abortions per year is estimated at 500000 (Malta et al., 2019). The prevalence of birth in the simulated dataset is 0.86, with a strong negative impact of zika virus, and a positive impact of socioeconomic status, see Table 2 for details.
- `hospital` ( $S_2$ ). Births that occurred in public hospitals were included in the study, and births that occurred in private hospitals were excluded. A majority of the population in Brazil visits public hospitals, and here it is assumed that it is strongly affected by socioeconomic status and weakly affected by living area, see Table 2 for details.



**Figure 2:** Selection process and causal structure for the simulated dataset. Left: flow chart of the two-step selection from the total population. Pregnancies ending in termination are excluded first, followed by births occurring at private hospitals, leaving the subpopulation with  $I_S = 1$ . Right: the corresponding causal DAG, where living area ( $V$ ) and socioeconomic status ( $U$ ) are unmeasured variables affecting the two selection steps, birth ( $S_1$ ) and public hospital ( $S_2$ ), while zika infection ( $T$ ) affects microcephaly ( $Y$ ) directly. The DAG shows that both selection steps depend on the same unmeasured variables that also relate to the outcome, which is why selection bias can arise even though  $T$  and  $Y$  are not confounded in the total population.

**Table 2:** Data generating process for the dataset zika\_learner. Models generating causal dependencies are logistic,  $g(X'\theta)$ , for predictor variables  $X$  (including a constant 1 for the intercept) and model parameter  $\theta$  (including the intercept).

| Model                                           | Coefficients ( $\theta$ )/Proportions | Function argument |
|-------------------------------------------------|---------------------------------------|-------------------|
| $P(V = 1)$                                      | 0.85                                  | Vval              |
| $P(U = 1)$                                      | 0.50                                  | Uval              |
| $P(T = 1 V) = g(V'\theta_T)$                    | $(-6.20, 1.75)$                       | Tcoef             |
| $P(Y = 1 T, U) = g[(T, U)'\theta_Y]$            | $(-5.20, 5.00, -1.00)$                | Ycoef             |
| $P(S_1 = 1 V, U, T) = g[(V, U, T)'\theta_{S1}]$ | $(1.20, 0.00, 2.00, -4.00)$           | Scoef             |
| $P(S_2 = 1 V, U, T) = g[(V, U, T)'\theta_{S2}]$ | $(2.20, 0.50, -2.75, 0.00)$           | Scoef             |

The unobserved variables, urban ( $V$ ) and ses ( $U$ ) are constructed such that the true sensitivity parameters in our DGP approximately match the assumed values in [Smith and VanderWeele \(2019\)](#). In the simulated dataset, the proportions of the variables presented in Tables 3-5 reflect this. The proportions are calculated both by treatment status (zika virus infection) and overall, for the total dataset, the subset with  $S_1 = 1$  and the subset with  $S_1 = 1$  and  $S_2 = 1$ .

**Table 3:** Proportions and number of subjects for the simulated dataset, by treatment (zika) status and overall.

|                    | Not zika infected | Zika infected | Overall |
|--------------------|-------------------|---------------|---------|
| Number of subjects | 4939              | 61            | 5000    |
| Microcephaly       | 0.003             | 0.361         | 0.008   |
| Living area        | 0.849             | 0.951         | 0.850   |
| SES                | 0.499             | 0.426         | 0.498   |

**Table 4:** Proportions and number of subjects for the simulated dataset, by treatment (zika) status and overall, after the first selection.

|                    | Not zika infected | Zika infected | Overall |
|--------------------|-------------------|---------------|---------|
| Number of subjects | 4268              | 11            | 4279    |
| Microcephaly       | 0.003             | 0.273         | 0.004   |
| Living area        | 0.845             | 1.000         | 0.846   |
| SES                | 0.556             | 0.818         | 0.557   |



For the DGP, the causal estimand is  $\beta_R = 90.7$  in the total population,  $\beta_{R_S} = 92.3$  in the subset of  $S_1 = 1$ , and  $\beta_{R_S} = 88.1$  in the subset of  $S_1 = 1$  and  $S_2 = 1$ . The observed risk ratios for the dataset are  $\beta_R^{obs} = 89.5$  (after one selection), and  $\beta_R^{obs} = 74.5$  (after two selections). The dataset and DGP can be used to apply the functions in [SelectionBias](#).

**Table 5:** Proportions and number of subjects for the simulated dataset, by treatment (zika) status and overall, after both selections.

|                    | Not zika infected | Zika infected | Overall |
|--------------------|-------------------|---------------|---------|
| Number of subjects | 2869              | 7             | 2876    |
| Microcephaly       | 0.004             | 0.286         | 0.005   |
| Living area        | 0.858             | 1.000         | 0.858   |
| SES                | 0.382             | 0.714         | 0.382   |

## 4 R package SelectionBias

In this section, we describe the package [SelectionBias](#). The different functions are illustrated using either the simulated dataset `zika_learner`, or the model structure that the dataset is simulated from (see Table 2). The package [SelectionBias](#) is available under a GPL-compatible license from the Comprehensive R Archive Network (CRAN) at <https://cran.r-project.org/package=SelectionBias>, and can be installed and loaded using:

```
install.packages("SelectionBias")
```

```
library(SelectionBias)
```

### 4.1 sensitivityparametersM()

The sensitivity parameters for the SV, sharp, and GAF bounds are calculated for the generalized M-structure illustrated in Figure 1 in the function `sensitivityparametersM()`. The input in the function is the model structure, and the observed probabilities of the outcome,  $P(Y = 1|T = t, I_S = 1)$ ,  $t = 0, 1$ . The code and the output are:

```
# SV bound
sensparSV = sensitivityparametersM(whichEst = "RR_tot",
  whichBound = "SV",
  Vval = matrix(c(1, 0, 0.85, 0.15), ncol = 2),
  Uval = matrix(c(1, 0, 0.5, 0.5), ncol = 2),
  Tcoef = c(-6.2, 1.75),
  Ycoef = c(-5.2, 5.0, -1.0),
  Scoef = matrix(c(1.2, 2.2, 0.0, 0.5,
    2.0, -2.75, -4.0, 0.0),
    ncol = 4),
  Mmodel = "L",
  pY1_T1_S1 = 0.286,
  pY1_T0_S1 = 0.004)

print(sensparSV)

#> $parameter
#> [1] "BF_11"      "BF_00"      "BF_10"      "BF_01"      "RR_UY|T=1" "RR_UY|T=0"
#> [7] "RR_SU|11"   "RR_SU|00"   "RR_SU|10"   "RR_SU|01"
#>
#> $value
#> [1] 1.2102 1.3532 1.3208 1.3895 1.9448 2.7089 1.5566 1.7058 1.9998 1.7998
#>
#> attr("class")
#> [1] "sensparams"
#> attr("row.names")
#> [1] 1 2 3 4 5 6 7 8 9 10
```

```

# Sharp bound
sensparSharp = sensitivityparametersM(whichEst = "RR_tot",
  whichBound = "sharp",
  Vval = matrix(c(1, 0, 0.85, 0.15), ncol = 2),
  Uval = matrix(c(1, 0, 0.5, 0.5), ncol = 2),
  Tcoef = c(-6.2, 1.75),
  Ycoef = c(-5.2, 5.0, -1.0),
  Scoef = matrix(c(1.2, 2.2, 0.0, 0.5,
    2.0, -2.75, -4.0, 0.0),
    ncol = 4),
  Mmodel = "L",
  pY1_T1_S1 = 0.286,
  pY1_T0_S1 = 0.004)

print(sensparSharp)

#> $parameter
#> [1] "BF_11"      "BF_00"      "BF_10"      "BF_01"      "RR_UY|T=1" "RR_UY|T=0"
#> [7] "RR_SU|11"   "RR_SU|00"   "RR_SU|10"   "RR_SU|01"
#>
#> $value
#> [1] 1.2102 1.3532 1.3208 1.3895 1.9448 2.7089 1.5566 1.7058 1.9998 1.7998
#>
#> attr("class")
#> [1] "sensparams"
#> attr("row.names")
#> [1] 1 2 3 4 5 6 7 8 9 10

# GAF bound
sensparGAF = sensitivityparametersM(whichEst = "RR_tot",
  whichBound = "GAF",
  Vval = matrix(c(1, 0, 0.85, 0.15), ncol = 2),
  Uval = matrix(c(1, 0, 0.5, 0.5), ncol = 2),
  Tcoef = c(-6.2, 1.75),
  Ycoef = c(-5.2, 5.0, -1.0),
  Scoef = matrix(c(1.2, 2.2, 0.0, 0.5,
    2.0, -2.75, -4.0, 0.0),
    ncol = 4),
  Mmodel = "L",
  pY1_T1_S1 = 0.286,
  pY1_T0_S1 = 0.004)

print(sensparGAF)

#> $parameter
#> [1] "m_T" "M_T"
#>
#> $value
#> [1] 0.0020 0.4502
#>
#> attr("class")
#> [1] "sensparams"
#> attr("row.names")
#> [1] 1 2

```

The first argument is `whichEst`, where the user inputs the causal estimand of interest. It must be one of the four "RR\_tot", "RD\_tot", "RR\_sub" or "RD\_sub". The second argument is `whichBound`, where the user inputs the bound they want to use, either "SV", "sharp", or "GAF". Third, the argument `Vval` takes the matrix for  $V$  as input. The first column contains the values that  $V$  can take, and the second column contains the corresponding probabilities. In this example,  $V$  is binary, so the first two elements in the matrix are 1 and 0. However, any discrete  $V$  can be used. An approximation of a continuous  $V$  can be used, if it is discretized. The fourth argument is `Uval`, which takes the matrix for  $U$  as input. The matrix  $U$  has a similar structure as  $V$ . The fifth argument is `Tcoef`, containing the coefficients used in the model for  $T$ . The first entry in `Tcoef` is the intercept of the model, and the second the slope for  $V$ .



The sixth argument is `Ycoef`, containing the coefficient vector for the outcome model, where the first entry is the intercept, the second the slope coefficient for  $T$  and third is the slope coefficient for  $U$ . The seventh argument is `Scoef`, the coefficient matrix for the selection variables. The number of rows is equal to the number of selection variables, and the number of columns is equal to four. The columns represent the intercept, and slope coefficients for  $V$ ,  $U$  and  $T$ , respectively. A summary of the code notation is seen in the last column of Table 2. The eighth argument is `Mmodel`, which indicates whether the models in the  $M$ -structure are probit (`Mmodel = "P"`) or logit (`Mmodel = "L"`). The ninth and tenth arguments are `pY1_T1_S1` and `pY1_T0_S1`. They are the observed probabilities  $P(Y = 1|T = 1, I_S = 1)$  and  $P(Y = 1|T = 0, I_S = 1)$ . The output is the sensitivity parameters for the chosen bound.

In the zika example, the estimand of interest is the risk ratio in the total population, `whichEst = "RR_tot"`, the DGP is found in Table 2, logistic models are used in the DGP, and the probabilities are found in Table 5. For the SV and sharp bounds, the output is  $RR_{UY|T=1} = 1.9448$ ,  $RR_{UY|T=0} = 2.7089$ ,  $RR_{SU|11} = 1.5566$ ,  $RR_{SU|00} = 1.7058$ ,  $RR_{SU|10} = 1.9998$ , and  $RR_{SU|01} = 1.7998$ , which gives  $BF_{11} = 1.2102$  and  $BF_{00} = 1.3532$ ,  $BF_{10} = 1.3208$ , and  $BF_{01} = 1.3895$ . For the GAF bound, the output is  $M_T = 0.4502$  and  $m_T = 0.002$ .

## 4.2 SVbound()

The SV bound can be calculated using the function `SVbound()`. The first argument is `whichEst`, indicating the causal estimand of interest ("`RR_tot`", "`RD_tot`", "`RR_sub`" or "`RD_sub`"). The second argument is `sens`, which is either output from `sensitivityparametersM()`, a named list, or a data frame with column names `parameter` and `value`, which includes the sensitivity parameters. This input is optional as the sensitivity parameters can also be specified directly in the function with the below arguments. The third and fourth arguments are the observed conditional probabilities  $P(Y = 1|T = 1, I_S = 1)$  and  $P(Y = 1|T = 0, I_S = 1)$ , which are needed to calculate bounds for the causal estimands and not the selection bias itself. The optional fifth and sixth arguments are the treatment probabilities  $P(T = 1|I_S = 1)$  and  $P(T = 0|I_S = 1)$ . These are only needed for the alternative SV bound for the causal risk difference in the subpopulation (Zetterstrom et al., 2025). If they are not included, the original SV bound for the risk difference is calculated. The subsequent arguments are the sensitivity parameters provided by the user. The default value for all sensitivity parameters is `NULL`, and the user must then specify numeric values on the sensitivity parameters that are necessary for the bound for the chosen estimand. These arguments are also optional in case the argument `sens` is used. The sensitivity parameters can either be calculated using the function `sensitivityparametersM()`, or found elsewhere. For sensitivity parameters found elsewhere, `SVbound()` is not restricted to the generalized  $M$ -structure. However, the necessary assumptions for the SV bound must still be fulfilled (Smith and VanderWeele, 2019). The output is the SV bound. The code and output are:

```
SVbound(whichEst = "RR_tot",
        pY1_T1_S1 = 0.286,
        pY1_T0_S1 = 0.004,
        RR_UY_T1 = 1.9448,
        RR_UY_T0 = 2.7089,
        RR_SU_11 = 1.5566,
        RR_SU_00 = 1.7058,
        RR_SU_10 = 1.9998,
        RR_SU_01 = 1.7998)

#>      [,1]      [,2]
#> [1,] "SV lower bound" 43.66
#> [2,] "SV upper bound" 131.22
```

As before in the zika example, the causal estimand is the risk ratio in the total population, `whichEst = "RR_tot"`. The sensitivity parameters are  $RR_{UY|T=1} = 1.9448$ ,  $RR_{UY|T=0} = 2.7089$ ,  $RR_{SU|11} = 1.5566$ ,  $RR_{SU|00} = 1.7058$ ,  $RR_{SU|10} = 1.9998$ , and  $RR_{SU|01} = 1.7998$ , calculated above in `sensitivityparametersM()`, which gives a lower SV bound equal to 43.66 and a higher SV bound equal to 131.22. Alternatively, the output from the function `sensitivityparametersM()` can be used directly as input which gives the code:

```
SVbound(whichEst = "RR_tot",
        sens = sensparSV,
        pY1_T1_S1 = 0.286,
        pY1_T0_S1 = 0.004)
```

```
#>      [,1]      [,2]
#> [1,] "SV lower bound" 43.66
#> [2,] "SV upper bound" 131.22
```

This approach gives the same bounds as specifying the sensitivity parameters manually.

### 4.3 sharpbound()

The sharp bound can be calculated using the function `sharpbound()`. The input arguments are very similar to that of `SVbound()`. The first argument is `whichEst`, indicating the causal estimand of interest ("RR\_tot", "RD\_tot", "RR\_sub" or "RD\_sub"). The second argument is `sens`, which is either output from `sensitivityparametersM()`, a named list, or a data frame with column names parameter and value, which includes the sensitivity parameters. This input is optional as the sensitivity parameters can also be specified directly in the function with the below arguments. The third and fourth arguments are the observed conditional probabilities  $P(Y = 1|T = 1, I_S = 1)$  and  $P(Y = 1|T = 0, I_S = 1)$ , which are needed to calculate bounds for the causal estimands and not the selection bias itself. The fifth and sixth arguments are the treatment probabilities  $P(T = 1|I_S = 1)$  and  $P(T = 0|I_S = 1)$ . These are only needed for the bounds for the causal estimands in the subpopulation. The seventh and eighth arguments are the probabilities  $P(I_S = 1|T = 1)$  and  $P(I_S = 1|T = 0)$  which are only needed for the bounds for the causal estimands in the total population. If they are unknown, they can be set to 0 for more conservative bounds. The subsequent arguments are the sensitivity parameters provided by the user. The default value for all sensitivity parameters is NULL, and the user must then specify numeric values on the sensitivity parameters that are necessary for the bound for the chosen estimand. These arguments are also optional in case the argument `sens` is used. The sensitivity parameters can either be calculated using the function `sensitivityparametersM()`, or found elsewhere. For sensitivity parameters found elsewhere, `sharpbound()` is not restricted to the generalized M-structure. However, the necessary assumptions for the sharp bound must still be fulfilled (Zetterstrom et al., 2025). The output is the sharp bound. The code and output with the two options to specify the sensitivity parameters are:

```
sharpbound(whichEst = "RR_tot",
            pY1_T1_S1 = 0.286,
            pY1_T0_S1 = 0.004,
            pS1_T1 = 0.11,
            pS1_T0 = 0.58,
            RR_UY_T1 = 1.9448,
            RR_UY_T0 = 2.7089,
            RR_SU_11 = 1.5566,
            RR_SU_00 = 1.7058,
            RR_SU_10 = 1.9998,
            RR_SU_01 = 1.7998)

#>      [,1]      [,2]
#> [1,] "Sharp lower bound" 73.91
#> [2,] "Sharp upper bound" 78.99

sharpbound(whichEst = "RR_tot",
            sens = sensparSharp,
            pY1_T1_S1 = 0.286,
            pY1_T0_S1 = 0.004,
            pS1_T1 = 0.11,
            pS1_T0 = 0.58)

#>      [,1]      [,2]
#> [1,] "Sharp lower bound" 73.91
#> [2,] "Sharp upper bound" 78.99
```

As before in the zika example, the causal estimand is the risk ratio in the total population, `whichEst = "RR_tot"`. The observed probabilities come from the dataset, and the sensitivity parameters are  $RR_{UY|T=1} = 1.9448$ ,  $RR_{UY|T=0} = 2.7089$ ,  $RR_{SU|11} = 1.5566$ ,  $RR_{SU|00} = 1.7058$ ,  $RR_{SU|10} = 1.9998$ , and  $RR_{SU|01} = 1.7998$ , calculated above in `sensitivityparametersM()`, which gives a lower sharp bound equal to 73.91 and a higher sharp bound equal to 78.99.

#### 4.4 checksharpSVbound()

The sharpness of the SV bound can be evaluated using `checksharpSVbound()` (Zetterstrom et al., 2025). The first argument is `whichEst`, indicating the causal estimand of interest ("RR\_tot", "RR\_sub" or "RD\_sub"). Note that the SV bound for the risk difference in the total population is not sharp. The second argument is `sens`, which is either output from `sensitivityparametersM()`, a named list or a data frame with column names `parameter` and `value`, which includes the bounding factors ( $BF$ ). This input is optional as the bounding factors can also be specified directly in the function with the argument below. The third argument is `BF`, the bounding factors  $BF_{00}$  and  $BF_{10}$  for the total population, and  $BF_0$  and  $BF_1$  for the subpopulation. The fourth argument is `pY1`, the probabilities  $P(Y = 1|T = 1, I_S = 1)$  and  $P(Y = 1|T = 0, I_S = 1)$ . Note that the order of the bounding factors and probabilities matters. The output is two strings stating whether the lower and upper SV bound are sharp or not. Note that the SV bound for the risk ratio in the total population and risk difference in the subpopulation can only be arbitrarily sharp, see (Zetterstrom et al., 2025) for more details. The code is:

```
checksharpSVbound(whichEst = "RR_tot",
  sens = sensparSV,
  pY1 = c(0.286, 0.004))

#> The lower SV bound for the risk ratio in the total population is arbitrarily sharp.
#> See vignette for details.

#> The upper SV bound for the risk ratio in the total population is arbitrarily sharp.
#> See vignette for details.
```

The output is a string that states whether the SV bounds are sharp or not. For these settings, both the lower and upper SV bounds are arbitrarily sharp.

#### 4.5 AFbound()

The AF bound is calculated using the function `AFbound()`. The first argument is the causal estimand of interest ("RR\_tot", "RD\_tot", "RR\_sub" or "RD\_sub"). The second argument is `outcome`, where the user inputs either the numeric vector with the observed outcome variable or a vector with the conditional outcome probabilities,  $P(Y = 1|T = 1, I_S = 1)$  and  $P(Y = 1|T = 0, I_S = 1)$ . The third argument is `treatment`, where the user inputs either the numeric vector with the observed treatment variable or a vector with the conditional treatment probabilities,  $P(T = 1|I_S = 1)$  and  $P(T = 0|I_S = 1)$ . The fourth argument is `selection` where the user inputs either the numeric vector with the observed selection variable, or the selection probability,  $P(I_S = 1)$ . Its default value is `NULL` since it is only required when the causal estimands in the total population are of interest. If the subpopulation is of interest and `selection = NULL`, the outcome and treatment vectors must only include the selected subjects. The output is the lower and upper AF bounds. The code and output are:

```
attach(zika_learner)

AFbound(whichEst = "RR_tot",
  outcome = mic_ceph[sel_ind == 1],
  treatment = zika[sel_ind == 1],
  selection = mean(sel_ind))

#>      [,1]      [,2]
#> [1,] "AF lower bound" 0
#> [2,] "AF upper bound" 454.09
```

Similar to before, `whichEst = "RR_tot"`. Furthermore, the outcome and treatment variables are microcephaly and zika. The selection probability is specified since the other variables are restricted to those subjects with  $I_S = 1$ . The output is the lower and upper AF bounds, which are 0 and 454.09 in the zika example.

If the raw data is not available, one can input the conditional probabilities instead. In this example, these probabilities are:

```
AFbound(whichEst = "RR_tot",
  outcome = c(0.286, 0.004),
  treatment = c(0.002, 0.998),
  selection = mean(sel_ind))
```

```
#>      [,1]      [,2]
#> [1,] "AF lower bound" 0
#> [2,] "AF upper bound" 435.14
```

The difference in these two examples comes from rounding errors in the input probabilities.

#### 4.6 GAFbound()

The GAF bound is calculated using the function `GAFbound()`. The first argument is the causal estimand of interest ("RR\_tot", "RD\_tot", "RR\_sub" or "RD\_sub"). The second argument is `sens`, which is either output from `sensitivityparametersM()`, a named list, or a data frame with column names "parameter" and "value", which includes the sensitivity parameters. This input is optional as the sensitivity parameters can also be specified directly in the function with the below arguments. The third and fourth arguments are `M` and `m` which are the two sensitivity parameters for the GAF bound. The sensitivity parameters can either be calculated using `sensitivityparametersM()`, or found elsewhere. For sensitivity parameters found elsewhere, `GAFbound()` is not restricted to the generalized M-structure. However, the necessary assumptions for the GAF bound must still be fulfilled (Zetterstrom, 2024). The fifth argument is `outcome`, where the user inputs either the observed numeric vector with the outcome variable or a vector with the conditional outcome probabilities,  $P(Y = 1|T = 1, I_S = 1)$  and  $P(Y = 1|T = 0, I_S = 1)$ . The sixth argument is `treatment`, where the user inputs either the observed numeric vector with the treatment variable or a vector with the conditional treatment probabilities,  $P(T = 1|I_S = 1)$  and  $P(T = 0|I_S = 1)$ . The seventh argument is `selection` where the user can either input the observed selection vector or selection probability. Its default value is `NULL` since it is only required when the causal estimands in the total population are of interest. If the subpopulation is of interest and `selection = NULL`, the outcome and treatment vectors must only include the selected subjects. The output is the lower and upper GAF bounds. The code and output are:

```
GAFbound(whichEst = "RR_tot",
         M = 0.4502,
         m = 0.002,
         outcome = mic_ceph[sel_ind == 1],
         treatment = zika[sel_ind == 1],
         selection = mean(sel_ind))

#>      [,1]      [,2]
#> [1,] "GAF lower bound" 0.01
#> [2,] "GAF upper bound" 147.42

GAFbound(whichEst = "RR_tot",
         sens = sensparGAF,
         outcome = mic_ceph[sel_ind == 1],
         treatment = zika[sel_ind == 1],
         selection = mean(sel_ind))

#>      [,1]      [,2]
#> [1,] "GAF lower bound" 0.01
#> [2,] "GAF upper bound" 147.42
```

Similar to before, the estimand of interest is the risk ratio in the total population, `whichEst = "RR_tot"`. The sensitivity parameters are the output from `sensitivityparametersM()`. Furthermore, the outcome and treatment variables are microcephaly and zika. The selection probability is specified since the other variables are restricted to those subjects with  $I_S = 1$ . The output is the lower and upper GAF bounds, which are 0.01 and 147.42 in the zika example. If the raw data is not available, one can input the conditional probabilities instead, similar to the AF bound.

#### 4.7 CAFbound()

The CAF bound is calculated using the function `CAFbound()`. The first argument is the causal estimand of interest ("RR\_tot", "RD\_tot", "RR\_sub" or "RD\_sub"). The second and third arguments are `M` and `m` which are the two sensitivity parameters for the CAF bound. The fourth argument is `outcome`, where the user inputs either the numeric vector with the observed outcome variable or a vector with the conditional outcome probabilities,  $P(Y = 1|T = 1, I_S = 1)$  and  $P(Y = 1|T = 0, I_S = 1)$ . The fifth

argument is treatment, where the user inputs either the numeric vector with the observed treatment variable or a vector with the conditional treatment probabilities,  $P(T = 1|I_S = 1)$  and  $P(T = 0|I_S = 1)$ . The sixth argument is selection where the user inputs either the numeric vector with the observed selection variable, or the selection probability,  $P(I_S = 1)$ . Its default value is NULL since it is only required when the causal estimands in the total population are of interest. If the subpopulation is of interest and selection = NULL, the outcome and treatment vectors must only include the selected subjects. The output is the lower and upper CAF bounds. The code and output are:

```
CAFbound(whichEst = "RR_tot",
         M = 0.3,
         m = 0.005,
         outcome = c(0.286, 0.004),
         treatment = c(0.002, 0.998),
         selection = mean(sel_ind))

#>      [,1]      [,2]
#> [1,] "CAF lower bound" 0.04
#> [2,] "CAF upper bound" 67.78
```

Similar to before, whichEst = "RR\_tot". The sensitivity parameters are chosen by the user. Furthermore, the outcome and treatment variables are microcephaly and zika. The selection probability is specified since the other variables are restricted to those subjects with  $I_S = 1$ . The output is the lower and upper CAF bounds, which are 0.04 and 67.78 in the zika example. If the raw data is not available, one can input the conditional probabilities instead, similar to the AF and GAF bounds.

## 5 Comparison to other software for sensitivity analysis

Sensitivity analysis is an important tool when the statistical analysis relies on untestable assumptions, and there are many R packages that can be used to perform sensitivity analyses for different types of bias and using different methods. Here, we address some of them and compare them to [SelectionBias](#).

The R packages [tipr](#) (McGowan, 2022) and [sensemakr](#) (Cinelli et al., 2024) solely focus on bias that arises from unmeasured confounding between a treatment and outcome variable (although [sensemakr](#) calls this "omitted-variable bias"), and they are therefore not concerning the same situation as [SelectionBias](#). The numerical bounds for multiple types of biases presented in Duarte et al. (2024) are accompanied by a Python package. These bounds make fewer assumptions than the SV, sharp, and GAF bounds, and can therefore be more conservative, and are not available in R. The R package with most overlap is the [EValue](#) package, and below follows a more detailed comparison of them.

The R package [EValue](#) provides sensitivity analysis methods for unmeasured confounding, measurement error, selection bias, and combinations of them (VanderWeele and Ding, 2017; Mathur and VanderWeele, 2020). However, [EValue](#) only provides one method of sensitivity analysis for selection bias: the SV bound (and other versions of the SV bound that require more information and assumptions) and the accompanying selection E-value. The selection E-value is a summary measure that calculates how large the sensitivity parameters must be in order to explain away an effect under the assumption that they all are equal, i.e. a type of Cornfield inequality (Cornfield et al., 1959).

[SelectionBias](#) on the other hand includes four other bounds (in addition to the SV bound): the sharp, GAF, CAF, and AF bounds. The user therefore has more options for their sensitivity analyses for selection bias. This is important since the different bounds are valid under different assumptions, meaning that they can be used in various cases. As a result, the package [SelectionBias](#) can cover more cases than the [EValue](#) package. The sensitivity parameters required in the SV bound may in some cases also be difficult to specify, and [SelectionBias](#) can calculate the sensitivity parameters for a user-specified DGP whereas [EValue](#) requires the user to specify them. This DGP may be easier to specify than the sensitivity parameters in some cases. [SelectionBias](#) can also check sharpness of the SV bound, i.e. check if it is possible that the bias is as large as the bound or if it is too conservative.

To summarize, if the assumptions for the SV bound are fulfilled, and the user is confident in specifying the sensitivity parameters for the SV bounds, or wants to calculate bounds for other types of biases as well, the package [EValue](#) provides the desired functionality. On the other hand, if the assumptions for the SV bound are not valid, or the user is more confident in specifying a DGP, or wishes to compare different bounds for the selection bias, [SelectionBias](#) provides the required functionality. Including several bounds in one framework, together with more comprehensive numerical and data examples, allows [SelectionBias](#) to facilitate comparisons between bounds. The different features of these packages are summarized in Table 6.

**Table 6:** Summary of the features of the different packages.

| Package       | Calculate sensitivity parameters from DGP | Multiple bounds                   | Additional SV bounds | Summary measure | Evaluate sharpness | Other biases |
|---------------|-------------------------------------------|-----------------------------------|----------------------|-----------------|--------------------|--------------|
| EValue        | No                                        | No                                | Yes                  | Yes             | No                 | Yes          |
| SelectionBias | Yes                                       | Yes:<br>sharp<br>GAF<br>CAF<br>AF | No                   | No              | Yes                | No           |

## 6 Conclusion

To perform sensitivity analyses for selection bias, an option is to calculate bounds. For applied researchers, we present the R package **SelectionBias**. The **SelectionBias** package provides functions for calculating sensitivity parameters and bounds previously proposed in the literature. In detail, the package includes a function that calculates the sensitivity parameters for the SV, sharp, and GAF bounds for the generalized M-structure specified by the user, functions for calculating the SV, sharp, AF, GAF, and CAF bounds, and a function for checking whether the calculated SV bound is sharp (Smith and VanderWeele, 2019; Zetterstrom et al., 2025; Zetterstrom, 2024; Zetterstrom and Waernbaum, 2022). For users to become familiar with the functions, the R package also includes a simulated dataset, `zika_learner`, and the DGP the dataset is simulated from. The simulated data emulates a register-based cohort study, inspired by a case-control study investigating the effect of zika virus on microcephaly in Brazil (de Araújo et al., 2018). Hopefully, **SelectionBias** can be a complement to the already existing package **EValue**. **EValue** is a good option when the user has knowledge of the sensitivity parameters, or when the user wants to address several types of biases. **SelectionBias** on the other hand can be an option if the user wants to compare several different bounds or has knowledge about the DGP. Furthermore, with **SelectionBias**, the user can perform the calculations for the sensitivity analysis for many possible designs and different bounds, and thereby gain a broader knowledge of the possible magnitude of the selection bias. The package is continuously updated with new sensitivity analyses and additional functions. For instance, bootstrap is expected to be included to account for sampling variability.

## Appendix

### Underlying assumptions for the SV, sharp, and GAF bounds

The SV, sharp, and GAF bounds are valid under additional assumptions. The bounds are derived under two different conditional independence assumptions for the causal model, depending on whether the total or subpopulation estimands are of interest:

1. (Total population estimands  $\beta_R$  and  $\beta_D$ ) For some unmeasured variable(s)  $U$ :  $Y \perp I_S | (T = t, U = u)$ , for  $t = 0, 1$ .
2. (Subpopulation estimands  $\beta_{R_S}$  and  $\beta_{D_S}$ ) For some unmeasured variable(s)  $U$ :  $Y(t) \perp T | (I_S = 1, U = u)$ , for  $t = 0, 1$ .

The assumptions differ since their purpose is to provide conditions for unbiased estimation of the estimands of interest under selection if the unmeasured variable was observed. The difference arises from either i) a lack of generalization of a causal effect from a subpopulation to the total population, or ii) a violation of conditional exchangeability in the subpopulation, i.e.,  $Y(t) \perp T | X, I_S = 1$  does not hold. There are however structures in which both assumptions are fulfilled, for example the generalized M-structure in Figure 1.

### The bounds

All bounds have the structure

$$LB < \beta < UB.$$

The SV bounds for the risk ratio in the total population are

$$\frac{\beta_R^{obs}}{BF_{11}BF_{00}} \leq \beta_R \leq \beta_R^{obs} \cdot BF_{10}BF_{01}.$$

The SV bounds for the risk difference in the total population are

$$\begin{aligned} \beta_D^{obs} - (BF_{11} - \frac{P(Y=1|T=1, I_S=1)}{BF_{11}} + P(Y=1|T=0, I_S=1) \cdot BF_{00}) \leq \beta_D \leq \\ \beta_D^{obs} + (BF_{01} - \frac{P(Y=1|T=0, I_S=1)}{BF_{01}} + P(Y=1|T=1, I_S=1) \cdot BF_{10}). \end{aligned}$$

The SV bounds for the risk ratio in the subpopulation are

$$\frac{\beta_R^{obs}}{BF_1} \leq \beta_{R_s} \leq \beta_R^{obs} \cdot BF_0.$$

The SV bounds for the risk difference in the subpopulation are

$$\begin{aligned} \beta_D^{obs} - \max(P(Y=1|T=0, I_S=1) \cdot (BF_1 - 1), P(Y=1|T=1, I_S=1) \cdot (1 - 1/BF_1)) \leq \beta_{D_s} \leq \\ \beta_D^{obs} + \max(P(Y=1|T=1, I_S=1) \cdot (BF_0 - 1), P(Y=1|T=0, I_S=1) \cdot (1 - 1/BF_0)). \end{aligned}$$

The alternative SV bounds for the risk difference in the subpopulation are

$$\begin{aligned} \beta_D^{obs} - [P(Y=1|T=0, I_S=1) \cdot P(T=1|I_S=1) \cdot (BF_1 - 1) + P(Y=1|T=1, I_S=1) \cdot P(T=0|I_S=1) \cdot (1 - 1/BF_1)] \\ \leq \beta_{D_s} \leq \\ \beta_D^{obs} + [P(Y=1|T=1, I_S=1) \cdot P(T=0|I_S=1) \cdot (BF_0 - 1) + P(Y=1|T=0, I_S=1) \cdot P(T=1|I_S=1) \cdot (1 - 1/BF_0)]. \end{aligned}$$

The sharp bounds are presented as bounds for the potential outcome probabilities which can then be used to construct bounds for the causal estimands. The bounds for the total population are

$$l_t = p(Y=1|T=t, I_S=1)\{p(I_S=1|T=t) + p(I_S=0|T=t)/BF_{11}\}$$

and

$$u_t = p(Y=1|T=t, I_S=1) \times [p(I_S=1|T=t) + p(I_S=0|T=t) \times \min(BF_{10}, 1/p(Y=1|T=t, I_S=1))],$$

which results in the following bounds for  $p(Y(t)=1)$ :

$$l_t \leq p(Y(t)=1) \leq u_t.$$

Upper (lower) bounds for the causal risk ratio and causal risk difference are found by combining  $l_0$  ( $l_1$ ) and  $u_1$  ( $u_0$ ). The sharp bounds for the subpopulation potential outcome probabilities are

$$l'_t = p(Y=1|T=t, I_S=1)[p(T=t|I_S=1) + p(T=1-t|I_S=1)/BF_t]$$

and

$$u'_t = p(Y=1|T=t, I_S=1) \times [p(T=t|I_S=1) + p(T=1-t|I_S=1) \times \min\{BF_{(1-t)}, 1/p(Y=1|T=t, I_S=1)\}],$$

which results in the following bounds for  $p(Y(t)=1|I_S=1)$ :

$$l'_t \leq p(Y(t)=1|I_S=1) \leq u'_t$$

which can then be used to construct bounds for the causal risk ratio and causal risk difference.

For the GAF, CAF, and AF bounds the potential outcome probabilities in the total population can be bounded as

$$\begin{aligned} P(Y=1, T=t, S=1) + [1 - P(T=t, S=1)] \cdot m_T < P(Y(t)=1) < \\ P(Y=1, T=t, S=1) + [1 - P(T=t, S=1)] \cdot M_T. \end{aligned}$$

The GAF bounds are calculated by using

$$m_T = \min_{t,u} P(Y=1|T=t, U=u)$$



and

$$M_T = \max_{t,u} P(Y = 1|T = t, U = u),$$

the CAF bounds are calculated by using

$$m_T = \min_t P(Y = 1|T = t, I_S = 0)$$

and

$$M_T = \max_t P(Y = 1|T = t, I_S = 0),$$

and the AF bounds are calculated by using  $m_T = 0$  and  $M_T = 1$ . These bounds for the potential outcome probabilities can then be used to construct bounds for the causal risk ratio and causal risk difference.

For the GAF, CAF, and AF bounds the potential outcome probabilities in the subpopulation can be bounded as

$$P(Y = 1, T = t|I_S = 1) + P(T = 1 - t|I_S = 1) \cdot m_S < P(Y(t) = 1|S = 1) < \\ P(Y = 1, T = t|S = 1) + P(T = 1 - t|I_S = 1) \cdot M_S.$$

For the GAF bounds,

$$m_S = \min_{t,u} P(Y = 1|T = t, U = u, S = 1)$$

and

$$M_S = \max_{t,u} P(Y = 1|T = t, U = u, S = 1),$$

for the CAF bounds

$$m_S = \min_t P(Y(t) = 1|T = 1 - t, I_S = 1)$$

and

$$M_S = \max_t P(Y(t) = 1|T = 1 - t, I_S = 1),$$

and for the AF bounds  $m_S = 0$  and  $M_S = 1$ . These bounds for the potential outcome probabilities can then be used to construct bounds for the causal risk ratio and causal risk difference.

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